

Scientific Reviewer

12+3 · WEIGHT 2 · substantial

Claim under review: in-the-wild distributed Franka teleop with unprecedented scene diversity improves DP success & OOD vs in-domain-only and vs OXE co-training.

Spine: Same Franka · many scenes · protocol > vibes · Fig.10 before Fig.8 · 2-16-8 reactive · steal protocol into BEHAVIOR

Accept as landmark dataset · punish “generalist” overclaim · Fig.10 before Fig.8

Claim under review

What authors must earn

Primary claim

In-the-wild distributed Franka teleop with unprecedented scene diversity improves diffusion-policy success and OOD robustness vs in-domain-only and vs OXE co-training.

Must hold

Table I scene gap · protocol transparency · Fig.10 matched-N · multi-env eval

Scope limit

Per-task co-trained specialists — not download-and-it-works generalists

Overclaim risk

Marketing “generalist policies” when experiments are specialists + tiny ID demos

Strengths — what earns Accept

Landmark dataset paper behaviors

- 1. Table I positioning** 564 scenes vs priors $\ll 100$; OXE agg ~ 311 but many lab-like
- 2. Operationalized diversity** Verbs, objects, scenes, viewpoints, interaction loci — not “we collected a lot”
- 3. Protocol transparency** GUI random tasks, augs, ~ 20 min scene cap $\rightarrow$ replicate process
- 4. Fig. 10 causal-ish** Matched 7362 traj, vary scene diversity — THE scientific highlight
- 5. Multi-environment eval** Lab / office / real household kitchen including Cook Lentils
- 6. Honest non-goal** No new architecture; builds on Diffusion Policy correctly
- 7. App. G calibration** 36k/24k refined sets with quality metrics — rare for dataset papers

Fig. 10 — cite BEFORE Fig. 8

Best causal-ish evidence for diversity

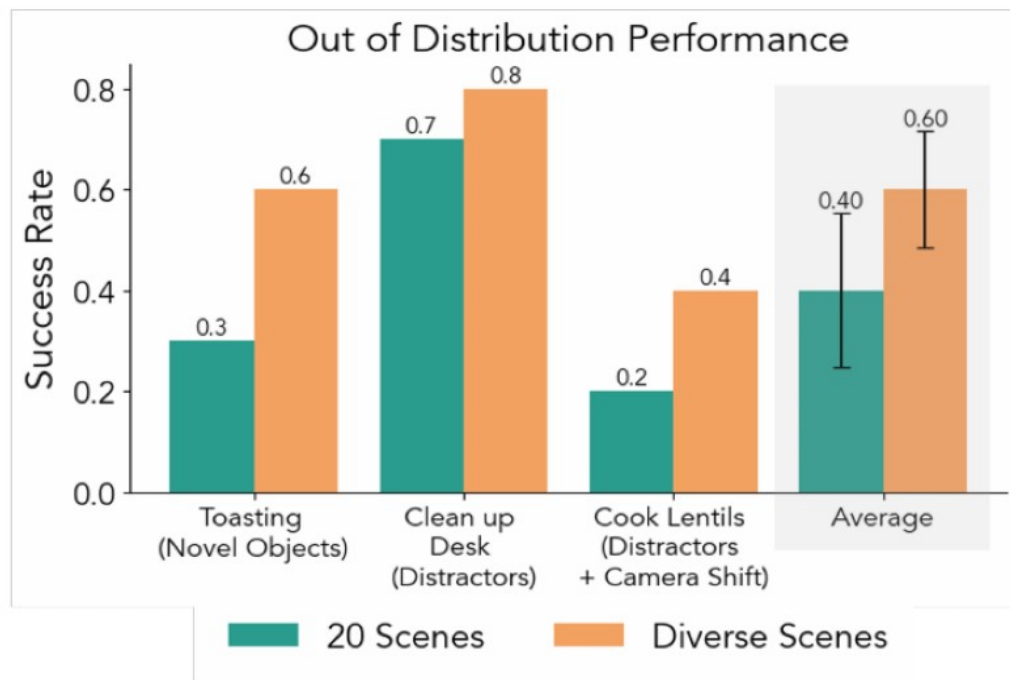


Fig. 10: **How important is the scene diversity in DROID?** We find that co-training on a subset of DROID with diverse scenes has higher OOD performance than co-training on a subset of DROID with only 20 scenes, suggesting the scene diversity of DROID is one of the

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Reviewer board

- Matched N = 7362
- Top-20 vs diverse scenes
- OOD 0.40 → 0.60 average
- Proves diversity $\neq$ hours
- Demand verb/object hist match too
- Collector-skill confound remains
- Still the paper's highlight

Fig. 8 — aggregate with caveats

+22% ID / +17% OOD · 10 rollouts/cell

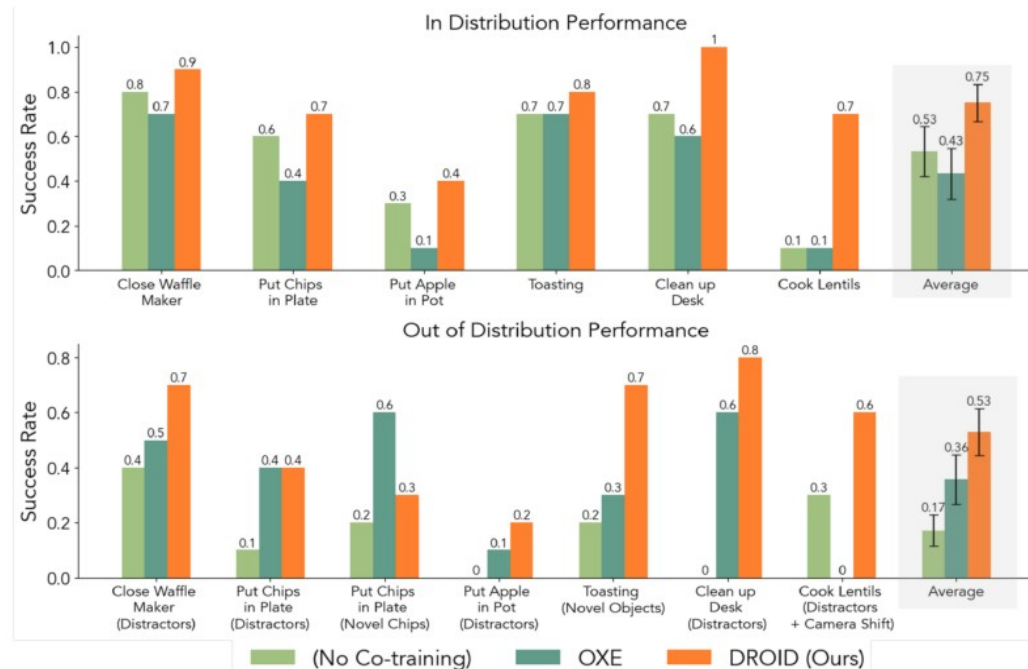


Fig. 8: Does DROID Improve Policy Performance and Robustness? We find that across all our evaluation tasks, co-training with DROID significantly improves both in distribution and OOD performance over both no co-training and co-training with the Open-X dataset. We compare success rate averaged across all tasks with standard error, and find DROID outperforms the next best method by **22%** absolute success rate in-distribution and **by 17%** out of distribution.

In Figure 10 we observe that using the split of the dataset with more diverse scenes yields better performance in the OOD evaluation setting. By comparing Figure 10's individual task performances with the corresponding tasks in Figure 8, we also see that the performance of co-training with the full DROID dataset matches or outperforms the performance with the subsampled dataset on all three tasks. These results suggest that the strength of DROID lies in its size and especially in

We hope that DROID can be a catalyst for research on general-purpose robot manipulation policies that are able to generalize to a broad range of tasks and scenes. In this work, we showed one example for leveraging DROID to boost policy performance, but there are many open questions about how to best make use of such diverse data: how should we combine DROID with existing large-scale robot datasets and how can we train policies that perform tasks in new scenes without any

Punish / demand

- SE bars; per-task tables light
- 10 rollouts/cell → want CIs
- Pre-register OOD defs
- ~40k co-train ≠ 76k headline
- OXE = Octo soup – Lang Table
- Not full 1.4M OXE
- Embodiment-match confound

Weaknesses — actionable (1/2)

What a good review demands

10 rollouts/cell

Demand CIs + pre-registered OOD definitions

40k $\neq$ 76k

Freeze versioned manifests for co-train pool

OXE soup $\neq$ 1.4M

Say curated Octo split; Language Table dropped

No 0-demo ZS

Authors admit open — punish “download and it works”

Wrist under-ablated

Selling point collected; paper policies ignore it

DP-only head

Cross-check Octo / VLA / π_0 before data-general claims

Weaknesses — actionable (2/2)

Confounds & ethics

OXE embodiment confound

Franka-match may drive gains — demand Franka-only
OXE slice

Fig. 10 residual confound

Top-20 may differ in task/object entropy / collector
skill

Scene dedup / GPT-4V

App. D heuristic; taxonomy needs human audit rates

Language IAA thin

Crowdsourced ≤ 3 instructions — report agreement

Long-horizon still chunks

No hierarchical planner / options baseline

Labor/safety externalities

In-the-wild collection underspecified for
replication ethics

OpenVLA drop $\neq$ refutation

Interaction effect, not bad data

Later sequel: OpenVLA mixes DROID @~10% then DROPS mid-train (action-token underfit).

Franka-DROID Wipe FT @15 Hz still wins vs DP/Octo — data remains useful under right action representation.

Hold

Scene diversity is a first-class lever; Fig.10 stands

Caveat

Diverse $\neq$ easy to fit under discrete AR tokens

Lesson

Capacity $\times$ action-rep $\times$ mix — co-design, don't blame shard

Verdict

Accept as landmark dataset paper

ACCEPT — landmark dataset

Scene-diversity ablation is the scientific highlight. Operationalized protocol + open stack + calibration upgrades.

PUNISH — “generalist” overclaim

Experiments = per-task co-trained specialists + tiny in-domain demos. No 0-demo ZS. Don't let marketing outrun Figure 10.

OpenVLA drop = interaction effect (capacity $\times$ action-rep), not a refutation of the dataset claim.

Reviewer exam bite

Fig.10 before Fig.8 · 40k vs 76k · Fig.6 only for loci

Cite Fig. 10 before Fig. 8 to prove diversity; cite 40k vs 76k for what was optimized; cite Fig. 6 only for interaction-location coverage (not main SR). Accept as landmark dataset; punish “generalist” overclaim when experiments are specialists.

Also ready

10 rollouts/cell · OXE soup not full 1.4M · no 0-demo ZS · wrist under-ablated · DP-only head · OXE embodiment confound

Numbers tattoo — pin these

Shared across roles · Nov 6 ready

76k

successful traj

350 h

teleop hours

564

scenes

52

buildings

86

task verbs

50

collectors

13 / 18

labs / robots

+22% ID

Fig. 8 lift

+17% OOD

Fig. 8 lift

7362

Fig. 10 matched N

2 / 16 / 8

obs/pred/act @15Hz

~40k ≠ 76k

co-train pool

Scientific Reviewer — close

Takeaways + exam bite

1. Accept as landmark dataset — Fig. 10 matched-N is the scientific highlight.
2. Punish generalist marketing; experiments are per-task co-trained specialists.
3. Demand: CIs, 40k $\neq$ 76k manifests, Franka-only OXE slice, wrist ablations, cross-head checks.
4. OpenVLA drop is an interaction effect — not evidence DROID is bad data.

EXAM BITE

Fig.10 before Fig.8; 40k $\neq$ 76k; Accept landmark + punish generalist overclaim.